

CIE-1 PYQ Bank: Model Answers

Handwritten Question Bank & Solutions

Syllabus Topics Covered:

Table of Contents

Out of CIE-1 Scope Questions (18 Questions)

p. 2

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The following questions were identified in past papers but fall outside the official CIE-1 syllabus. They are preserved here to ensure zero questions are dropped.

Paper / Origin	Question Summary	Exclusion Rationale
Ramaiah CIE-1 2026 (Portions L1-L13)	Consider dataset of 10 samples for predicting whether to Play Tennis. Compute information gain of attribute outlook using ID3 algorithm.	<u>Decision Tree Learning (ID3, Entropy, Information Gain) is out of CIE-1 scope. ML Unit 3 scope is strictly limited to Regression.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Explain the working of the CART algorithm.	<u>CART Decision Tree Induction is out of CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Apply AGNES Algorithm for distance matrix clustering.	<u>Hierarchical Clustering belongs to Unit 4, outside CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Apply Naive Bayes classification algorithm for Outlook/Play dataset.	<u>Bayesian Learning belongs to Unit 4, outside CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Explain the Elbow Method used for selecting the optimal value of k in clustering.	<u>K-Means Clustering belongs to Unit 4, outside CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Age and Loan predictors in dataset. Apply K-NN Regression to find House Price Index (HPI) for Age=48, Loan=\$142,000 for K=3.	<u>Similarity-based learning (K-NN regression) is excluded from CIE-1 scope.</u>

Paper / Origin	Question Summary	Exclusion Rationale
Ramaiah CIE-2 2026 (Portions L15-L28)	Illustrate with an example DBSCAN clustering algorithm: core, border, noise points.	<u>Density-based clustering belongs to Unit 4, outside CIE-1 scope.</u>
Ramaiah CIE-2 2026 (Portions L15-L28)	Explain the working of Random Forest and AdaBoost algorithms with suitable examples.	<u>Ensemble Learning belongs to Unit 5, outside CIE-1 scope.</u>
SEE June/July 2026	For the Play-Tennis dataset, build the complete ID3 Decision Tree showing Entropy and Information Gain at each level.	<u>Decision Tree ID3 is out of CIE-1 scope.</u>
SEE June/July 2026	Naive Bayes classifier for Flu prediction; Linear SVM boundary and support vectors for bank fraud.	<u>Bayesian Learning and Support Vector Machines belong to Units 4 & 5.</u>
SEE June/July 2026	K-Means clustering algorithm, DBSCAN core/border/noise, Elbow method for k.	<u>Clustering belongs to Unit 4.</u>
SEE June/July 2026	DIANA hierarchical clustering, Single-linkage clustering, Expectation Maximization (EM).	<u>Clustering & EM belong to Unit 4.</u>
SEE June/July 2026	Bagging vs Boosting, Feature subset selection, Model selection using hypothesis tests.	<u>Ensemble learning belongs to Unit 5.</u>
SEE July 2024	ID3 Decision Tree with Gain Ratio; Naive Bayes with Laplacian correction.	<u>Decision trees and Bayes classifiers are out of CIE-1 scope.</u>
SEE July 2024	K-Means, Agglomerative Hierarchical, Perceptron, Backpropagation in Multilayer Neural Networks.	<u>Clustering and Neural Networks belong to Units 4 & 5.</u>

Paper / Origin	Question Summary	Exclusion Rationale
SEE May 2023	Decision Tree C4.5 split info; KNN classification with distance weights.	<u>Similarity-based learning and Decision trees are excluded from CIE-1 scope.</u>
SEE May 2023	Naive Bayes document classification, SVM dual formulation, K-Means clustering, Backpropagation.	<u>Bayesian, SVM, and Neural Networks belong to Units 4 & 5.</u>
MAKEUP 2024 & SUPPLE 2025	Ensemble AdaBoost, Random Forests, K-Means, Perceptron algorithm.	<u>Ensembles, Neural Networks, and Clustering belong to Units 4 & 5.</u>